Alpha Spectroscopy

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Abstract

In this article, we studied the α-spectroscopy through the decay chain of ^{226}Ra source using the Si detector and estimated its activity and age. The measured age was 16.257$\pm$0.20 years, approximately 9σ away from the true age of 18.07 years. Moreover, the statistical precision was studied and was experimentally confirmed within 8% uncertainty for the second peak and 46 % for the full spectrum.

1 Introduction

The radioactive decay process is one of the most intriguing phenomena in the nature. It has been a hot topic since its discovery by Henri Becquerel and Marie Curie in the late 1800s. During radioactive decay, new atomic nuclei are formed from the decaying atom with the release of huge amount of energy along with some other exotic particles like α and β-particles. In nuclear physics, a radioactive cascade or a decay chain is the radioactive disintegration of a parent radioactive nuclide into its stable daughter nuclei with the formation of many unstable intermediate products in between. When this happens, the time it takes for the parent nuclei to decay into its stable daughter nuclei vary widely from atom to atom.

In this experiment, we are mainly focused on those decay chains which releases α particles during the decay process. When a radioactive nuclei undergoes α -decay, the atomic and mass number of the daughter nuclei changes respectively by two and four as a α particle is non other then helium nuclei[1].

$$ {}^{A}_{Z}X_N \longrightarrow {}^{A-4}_{Z-2}Y_{N-2} + {}^{4}_{2}He_2 \tag{1} $$

The decay rate, measured experimentally, and the half life of the parent nuclei, then, can be used to determine the age of parent nuclei.

2 Theory

2.1 Decay law

The radioactive disintegration of identical nuclei over time t follows the exponential decay law. Initially, if N_0 number of nuclei were present, then the number of nuclei, N, after time t is given by,

$$ N = N_0 e^{-t/\tau} \tag{2} $$

where τ is the mean lifetime of the decaying nuclei expressed as

$$ \tau = \frac{t_{1/2}}{ln2} \tag{3} $$

The $t_{1/2}$ is the half life of the decaying atom, meaning in that time half of the nuclei present in the parent nuclei is decayed. The half lives varies from nuclei to nuclei, which can be used to determine the decay constant. When the decay happens, it is accompanied by the release of other particles (α, β etc) or photons (X-rays, γ-rays).

2.2 Activity of Source

The rate at which the radioactive source decays differs from nuclei to nuclei. It is, therefore, obvious that a nuclei with shorter half-life decays faster than a nuclei with longer half-life. The decay rate of ^{226}Ra is simply the change in its number with time and can be expressed as:

$$ -\frac{dN_A}{dt} = \frac{N_A}{\tau_A} \tag{4} $$

The negative sign implies that as time passes on the number of nuclei keeps decreasing.

To find the activity of the source we proceed as follows: The efficiency $\mathcal{E}$ of the detector of active area S kept at a distance R from the source is given by

$$\mathcal{E} = \frac{S}{4\pi R^2} \tag{5}$$

If the detector detects N particles in an experimental run time t', then the activity $\mathcal{A}$ of the source is given by

$$\mathcal{A} = \frac{N}{\mathcal{E}} = \frac{4\pi N R^2}{t' S} \tag{6}$$

2.3 Age of the Source

The age t of the parent nuclei defined as the time since it started decaying from its pure state can be estimated by using the following relation [2]:

$$t = -\frac{\tau_A \tau_B}{(\tau_A - \tau_B)}\, ln\left(1 - R\frac{\tau_A - \tau_B}{\tau_A}\right) \tag{7}$$

with the ratio of the decay rate of nuclei ^{210}Pb (B) over ^{226}Ra (A) is ,

$$R = \frac{N_B \tau_A}{N_A \tau_B} \tag{8}$$

where τ_A and τ_B are the mean lifetime of nuclei A and B respectively.
Once we know the values of τ_A,τ_B,and R, we can use this information to extract the age of the parent nuclei.

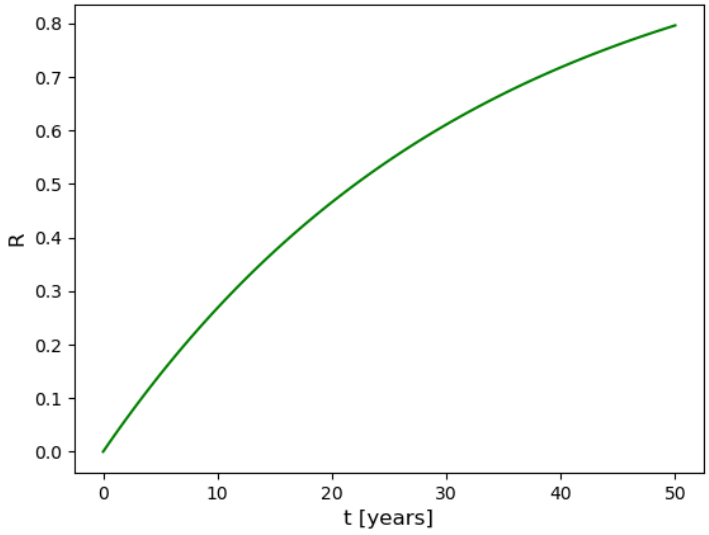


Figure 1: Ratio R of the decay rates of ^{210}Pb over ^{226}Ra as a function of time

2.4 Uncertainty

Every measurement we make are prone to error due to various reasons involved. Some errors like measurement error and human error can be controlled and improved over time taking large number of observations. However, there are some errors like the systematic error and apparatus error which persists even though we try our best to minimize them.

The precision in the measurement of N counts is given by the relation,

$$\sigma = \frac{1}{N-1}\sqrt{\sum_i (N_i - \bar{N})} \tag{9}$$

where N_i is the total sets of observation and $\bar{N}$ is the average of those observations. In principle, this statistical precision is expected to go as $\sqrt{N}$. The uncertainty in the measurement of the age of the source is calculated using the following equation:

$$\sigma_t = \left|\frac{\partial t}{\partial R}\sigma_R\right| = \frac{\tau_A \tau_B}{\tau_A(1-R) + \tau_B R}\sigma_R \tag{10}$$

with σ_R being the uncertainty in the measurement of the decay ratio.

3 Experimental Methods

The experimental apparatus we used in this experiment is shown in the schematic diagram below in Figure 1. We started by venting the evacuated chamber while the high voltage supply was turned off. This was achieved by closing the roughing pump off from the chamber and slowly opening the chamber valve. The valve dial was slowly turned with small steps to keep the change in pressure low so as not to damage the detector. The chamber was unscrewed once the atmospheric pressure was reached and the measurements were taken from the source-holder to the Silicon detector which are tabulated in Table 1.

Spec	Distance (cm)
Source to detector	8.6 ± 0.1
Detector Recession	0.3 ±0.1
Total Distance	8.9 ± 0.14

Table 1: Measured distance between source and detector with detector recession and uncertainty

The source, ^{226}Ra, was then positioned at the centre of the evacuated chamber, approximately, 8.9cm far from the detector. The dial handle and source face were aligned co-linearly and the chamber was re-screwed in the similar fashion to ensure the uniform gasket seal. A steady current less than $1\mu A$ was maintained throughout the experiment by applying a bias voltage of 90 V to the detector. The detector then detects the α particle and the signal from the detector was passed into an amplifier with the output split into a unimodal and bimodal signal.

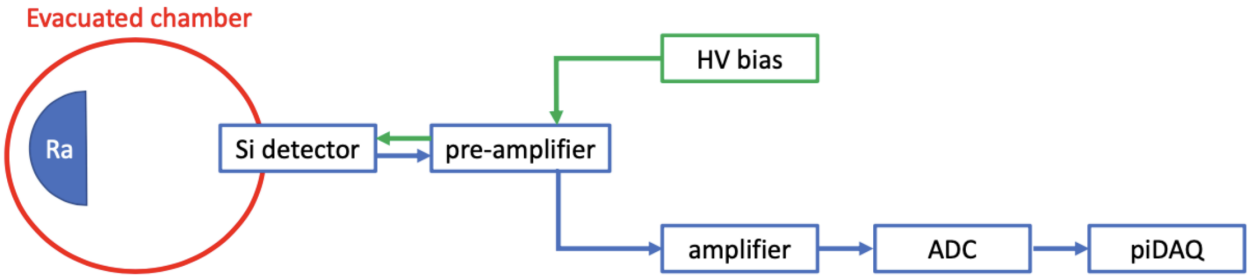


Figure 2: Experimental setup for the α spectroscopy experiment[2]

The unimodal signal was fed into a timing scaler which we used to strobe the PiDAQ ADC. The PiDAQ ADC is a 8091 bin ADC. The bimodal "energy" signal was fed into a delay amplifier before being routed into the PiDAQ ADC. The strobe signal and the bimodal energy signal were aligned using an oscilloscope and a delay amplifier. We conducted a total of 24 runs including 5 gain study and 7 to study the angular dependence.

Tag	Duration	Gain (Course)	Gain (Fine)	Gain (%)
tag0020	10	100	14	140
tag0021	N/A	100	15	150
tag0022	10	100	9	90
tag0023	5	50	9	45
tag0024	10	100	9	90

Table 2: Gain settings on the amplifier to determine the appropriate amplifier gain

Tag	Angle	Duration (s)
tag0030	0 ± 0.5	300
tag0031	0 ± 0.5	300
tag0032	20 ± 0.5	300
tag0033	40 ± 0.5	300
tag0034	60 ± 0.5	300
tag0035	90 ± 0.5	300

Table 3: Different runs used for angular dependence study

4 Analysis and Results

The spectra we got initially had an interesting structure where a steep cutoff at some maximal ADC bin was observed as shown in below Fig. 3. To ensure that there was nothing wrong with the amplifier and the ADC was not saturated by incorrect settings, we did several runs

at different amplifier gains. We observed that the bins at which the peaks occur shifted overall, but the peak themselves did not change.

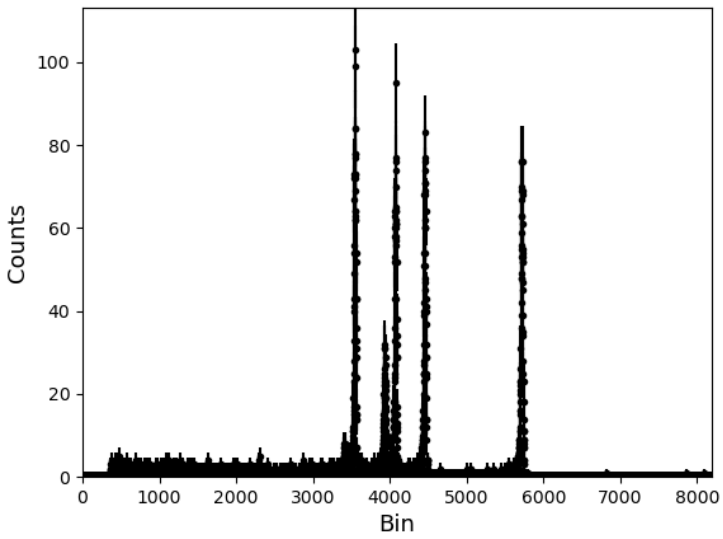


Figure 3: Spectrum showing a sharp cutoff at some maximal ADC bin

To study the dependence of decay rate on angle between the source and the detector, we took several short runs varying angle between them. The total rate was evaluated by taking the total count of the full spectrum and dividing by the total run time. The decay rate at different angles are tabulated below and we saw no abrupt change until 90° implying that the source emits the α-particle isotropically. Once we go beyond 90°, we saw a rapid drop in the decay rate which is reasonable as not all the emitted α-particles could make their way to the detector. Therefore, it is evident and wise to make the source face the detector for the best possible result. We expect five prominent peaks as there are mainly five α-decay

Angle	Rate (s^{-1})
0 ± 0.5	21.65 ± 0.27
0 ± 0.5	21.16 ± 0.27
20 ± 0.5	22.16 ± 0.27
40 ± 0.5	21.37 ± 0.27
60 ± 0.5	21.41 ± 0.27
90 ± 0.5	8.52 ± 0.17

Table 4: Decay rates at different angle between the source and the detector

chains between ^{226}Ra and ^{210}Po. However, we only need two peaks to calculate the ratio of decay rate and hence the age of parent nuclei and so we chose the first two peaks.

The raw data from the ADC were processed and was Gaussian fitted (see Fig. 4 and 5), the values of different parameters are then tabulated below. With these observed counts, the statistical precision in the measurement was also estimated.

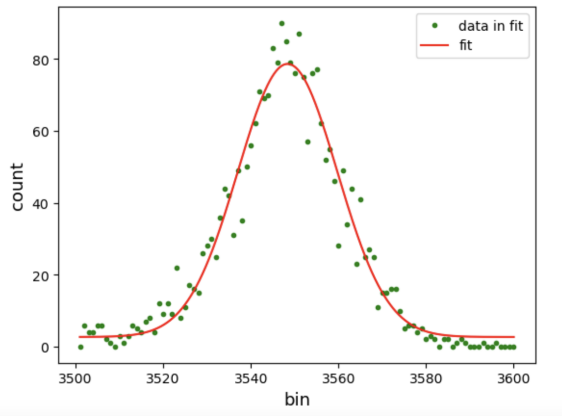


Figure 4: Gaussian fit for the first peak

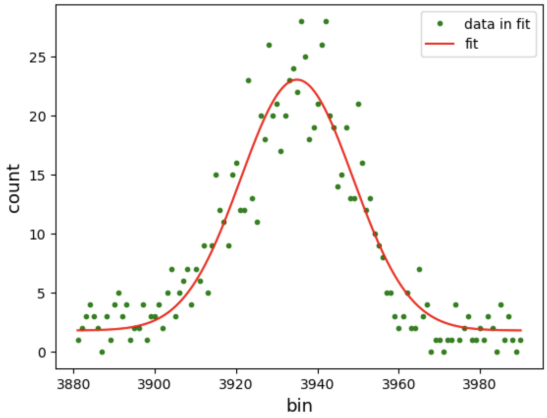


Figure 5: Gaussian fit for the second peak

Spec	First peak	Second peak	Full Spectrum
Mean counts	2444	976	13309
Expected precision	49	31	115
Observed precision	31	33	216

Table 5: The mean bins, counts and statistical precision of the first two peaks and whole spectrum

To calculate the age of the source, we first evaluated the parameters required, the decay rate of the two nuclei, their ratio and the results are tabulated below:

τ_A[years]	τ_B[years]	Activity [Bq]	R	Actual age [years]	Measured age [years]
2308.312	32.028	1351.508 ± 42.519	0.399	18.070	16.257±0.20

Table 6: Estimated age of the source

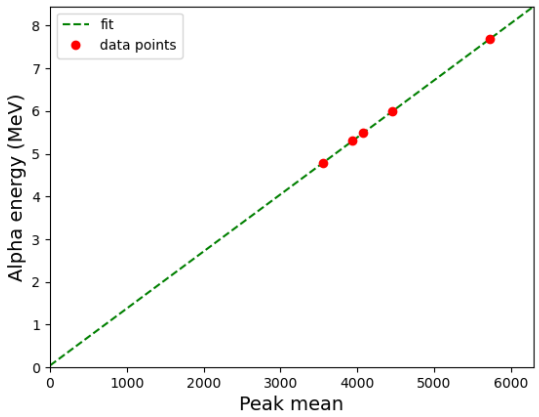


Figure 6: Energy calibration [3]

5 Conclusions

We studied the alpha decay spectrum through the decay chain of the ^{226}Ra source and a Silicon semiconducting detector in great detail as possible. Energy calibration was used taking standard value of α-decay energy to identify the peaks and corresponding energies. The energy calibration shows that our fit data matches exactly with the standard one with negligible residual energy. Various parameters were studied with different orientation of the source facing the detector. The age of the ^{226}Ra source was found to be 16.257 years, which is 9σ away from its actual age. The observed activity of source and its age were in reasonable agreement with the standard value within some error. The study of the statistical precision showed that it was in close agreement with the theoretical $\sqrt{N}$ within 8% for second peak . In future, we could go through a rigorous analysis of the potential systematic errors including the effects of physical defects of source and detector.

References

[1] K. Krane, *Introductory Nuclear Physics, 3rd Edition*(Wiley, 1987)

[2] G. Jensen, P.M King, and J. Roche, *PHYS6751: Nuclear Graduate Lab Manual v24.5* (Ohio University, 2024)

[3] National Nuclear Data Center, α-decay energy for various decay chains, https://www.nndc.bnl.gov/nudat/